

HealthBridge — Architecture Discussion Summary

This document summarizes the discussion regarding the future telemetry architecture for the HealthBridge Android family monitoring application.

Scenario #1 — Continuous Telemetry

Each phone continuously publishes GPS coordinates and other telemetry data (heart rate, battery level, etc.) to Firebase at fixed intervals (example: every 1–2 minutes).

Advantages: Simple architecture Reliable emergency monitoring Fast viewing of member locations Easy Firebase implementation Disadvantages: Higher battery usage Continuous location tracking raises privacy concerns More database writes as the system scales

Scenario #2 — On-Demand Requests

Phones stay mostly idle and periodically check whether another member has requested data. If requested, the phone sends fresh telemetry only to the authorized requester.

Advantages: Better privacy Lower bandwidth and database usage Lower battery consumption Disadvantages: Much more complex architecture Potential delays or failures due to Android sleep policies Less reliable for emergency situations

Recommended Hybrid Architecture

The recommended solution is a hybrid system: Low-frequency continuous telemetry (base layer) Optional live on-demand requests for detailed data Example: GPS every 2–5 minutes Battery every 5–10 minutes Immediate high-frequency updates only during alerts or emergencies This architecture provides: Reliability Emergency readiness Good battery efficiency Scalability for future features

Estimated Firebase Cost

At one update per minute: 1,440 writes/day/device Approximately 1 MB/day/device maximum Typical prototype usage is expected to remain within Firebase free-tier limits.

Estimated Battery Usage

Using high-accuracy GPS continuously: Approximately 5–15% battery/day Using balanced power accuracy: Approximately 2–6% battery/day Best practice: Use low-frequency updates normally Temporarily increase tracking frequency only during alerts or movement

Current Recommendation for HealthBridge

For version 1 of HealthBridge: Use continuous low-frequency telemetry Publish GPS every 2 minutes Use balanced power accuracy Add request-based live telemetry later The current multi-device Firebase and Google Maps architecture is now stable enough to support further development.